Pond, Manual

A pond for making music. Version 1.3.1.

Pond is a MIDI sequencer built on one idea: **drop a stone in the water and let the wave play the notes.** Every cell of the pond holds a note. When a ripple's wave front crosses a cell, that note sounds. Where you tap, what notes you've planted, and how the waves overlap, that's the composition.

---

1. Thirty seconds to first sound

> **First launch** runs a short interactive intro: tap **Drop some notes** to fill the water, then it walks you, one gentle step at a time, through tapping to play, planting an emitter and holding it to change its rhythm, painting, and filling the pond with **Generate**. Want it again later? Open the drawer and tap **Show intro again** in the About section at the bottom.

1. Open Pond. The pond starts **still and empty**, dark water, waiting.
2. Open the **drawer** (slider button, top-right) and tap **Generate**. The pond fills with a sparse scatter of notes.
3. **Tap anywhere.** A wave expands from your finger; every cell it crosses plays. Sound comes from the built-in sound, no setup needed.
4. Tap with **several fingers**. Each finger drops its own wave.
5. **Hold a finger still** for half a second. You've planted an **emitter**, a spring that keeps dropping ripples on its own, in time. Hold it again to open its menu: pick its rhythm, its MIDI channel, or **Remove** it. (Tapping or holding *on* an emitter's ring won't drop a stray ripple, the ring is a quiet spot.)

That's the whole instrument. Everything else shapes it.

---

2. Reading the pond

- **Each cell is a note.** Its colour tells you the pitch class, the palette runs from warm **peach (C)** through pink, orchid, and lavender to **sky blue (B)**. Cells brighten as their velocity rises.
- **Dark cells are silent** (a faint grey square at rest), and they **light up blue** as a wave front passes over them, so you can watch the wave travel across the pond.
- A lit cell **flashes** at the moment it plays; the moving band of lit cells *is* the wave, there's no separate ripple drawn over the grid.
- The grid is always made of perfect squares and fills the screen. The pond itself **stays put when you rotate** the iPad, like water in a tray, so every note and emitter keeps its place; only the controls swing around to the new orientation.
- The pond is **up to 32×24**. On a 12.9"/13" iPad (and in the plugin) you get the full 32×24; a narrower iPad (11"/Air/mini) shows a few rows fewer. A note or emitter on a row a given iPad doesn't show simply won't play there, so a pond made on a larger iPad may sound a little sparser on a smaller one. Generate only ever fills the rows you can see.

The **pill** in the top-left shows tempo · division · key at a glance. Tap it, or the slider button top-right, to open the **drawer**, where all the controls live. The **paintbrush button** just below the drawer button opens the **paint bar**, a floating palette that docks over the bottom of the pond. It glows orange while you're editing, and the bar itself is your "you're editing now" signal. Below the paintbrush sit **Generate** (the dice: deal a fresh pond of notes without opening the drawer) and **Clear** (the trash can: silence every note; emitters keep going). Both step back with Undo. The pill, the button column, and the pattern switcher **dim while your fingers are on the pond**, so they don't distract, and come back a second after you lift off. They still work while dimmed.

The **pause button** (top of the top-right buttons) freezes the pond like a sequencer's stop: waves halt in place and the sound cuts cleanly, with no hung notes. Tap it again to pick up exactly where you left off. (Inside an AUv3 host the host's own transport runs the show: the button is hidden, and the emitters start and stop with the host's play/stop instead, see *Pond as an AUv3 plugin*.)

3. Gestures

Gesture	In play mode	In edit mode
Tap	Drop a ripple	Paint the selected note
Drag	Move an emitter you're on	Paint continuously
Multiple fingers	One ripple per finger	Paint with each
Hold still (½ s)	Plant an emitter, or open its menu	Paint (same as a tap)

Edit mode is summoned by the **paintbrush button** (top-right, under the drawer button). The **paint bar** slides up over the bottom of the pond. Pick a pitch from the twelve chips, set the octave, or tap the **eraser** chip to paint silence. The chips for notes **in the current scale** keep their colour; out-of-scale notes dim to grey (still pickable), so the in-key choices stand out at a glance. Tap **Done** (or the paintbrush again) to put it away. The pond keeps playing the whole time.

The bar has a small **grab handle** on top: **drag or flick it** and it springs to the nearest resting spot, like a Picture-in-Picture video, so you can always clear the cells it was sitting over. It settles along the top or bottom, to the left, right, or back in the centre where it started.

Misplaced a note? Tap the same cell again, with the same note, within five seconds and it lifts back off, a quick undo without reaching for the eraser. For anything bigger, the drawer's **Undo** steps back whole edits.

4. The drawer, top to bottom

Every section opens expanded. **Tap a section's title to fold it away** (tap again to bring it back), so for a performance you can strip the drawer down to just the controls you play with. Pond remembers your layout between sessions.

Undo and Redo

Two buttons at the very top step backward and forward through your edits, up to ten of them, in the app and the plugin alike. An "edit" is anything that changes the pond itself: a paint stroke (a whole drag counts as one), Generate, Clear, Transpose, planting, moving, or clearing emitters, picking an emitter's rate or channel, switching patterns, or changing the scale, key, octave range, density, or the Physics controls. Tempo, quantize, MIDI routing, and volume are deliberately left out, Undo is for the notes, not the transport. History is per session and starts fresh each launch.

Presets (patterns)

Forty-eight slots across eight pages of six (flip with the ‹ › arrows or a swipe across the slots), each with a snapshot picture of its pond. Think of them as **patterns**: save a few pond variations and string them into a set.

- **Tap an empty slot** → saves the current pond there.
- **Tap a filled slot** → loads it instantly (waves in flight keep rolling over the new notes, lovely for transitions).
- **Long-press a filled slot** → overwrite with the current pond, rename, or delete.
- **Reorder** (the button above the slots) → the pages unfold into one list and you **drag patterns into the order you want**, so a progression plays in sequence. Tap Done when it reads right.

The pattern you're on wears a **blue ring**, and once you have any patterns saved a small **‹ name › switcher** floats at the bottom-left of the pond: tap the arrows to hot-switch to the previous or next pattern **without opening the menu**. Play a pond, switch, play the next, build a whole piece live. (The switcher tucks away while you're painting or when the menu is open, and undo steps a switch back.)

Each slot is tagged with the grid it was saved at, e.g. `Drift (32×24)`. If a preset's grid is taller than your iPad shows, its bottom rows just won't play here, the tag is your heads-up.

A preset stores the *musical* state: grid, emitters (with their rates and output channels), scale and key, tempo, division, quantize, drip rate, sustain, falloff, density. It deliberately does **not** touch your setup, built-in sound on/off, volume, and MIDI-clock settings stay as you have them.

Your patterns follow you into the plugin: the app and the AUv3 share one library, so a pattern saved in the app is right there when Pond runs inside AUM or Logic (and the other way round). Inside a DAW the host's tempo wins over a pattern's saved BPM, as always.

Pond also autosaves everything when you leave the app and restores it on return (including which pattern you were on), so you never lose the pond you were just playing.

Tempo

- **BPM** (20–300) and **Division**, together they set the wave speed: one cell per division step. At 120 BPM and 1/16, a wave crosses eight cells per second.
- **Quantize**, the pocket switch. Off, notes sound the instant the wave touches them (pure water physics, rubato by nature). Set to ½, ¼, ⅛, 1/16, 1/32, 1/64 or 1/128, every note instead waits for the next beat boundary: the waves stay liquid, the rhythm locks tight. The flash waits with the note, so what you see is what you hear.

MIDI

(Standalone only; inside an AUv3 host the host handles routing and clock.)

- **Send to**, choose which connected destinations receive Pond's output: a USB interface or synth, a Bluetooth or network device, or another app's input. The always-on "Pond" source still reaches other apps; this adds direct routing to the ones you tick. Notes go out on channel 1 unless you give an emitter its own channel (see *Emitters*). (More in *Pond and MIDI* below.)

- **Sync to MIDI clock**, Pond follows external MIDI clock (gear, AUM's clock, etc.). While ticks arrive the BPM shows "· EXT" and locks; it releases automatically when the clock stops. (Tempo matches; bar phase doesn't, for phase-locked sync use the AUv3.)
- **Send MIDI clock**, Pond becomes the clock master, emitting 24 ppqn from its MIDI output, tracking every BPM change.

Emitters

- **Per-emitter drip rate.** Each emitter keeps its OWN rate, so several emitters together make polyrhythms. **Hold an emitter** to open its menu and pick a rate directly: the bar multiples (8/4/3/2/1), or ½, ¼ and ⅛ each in **straight, dotted, or triplet** feel, so you can build shuffles and cross-rhythms. (Picking directly means you never cycle *through* the fast rates to reach a slow one.) Each emitter's ring marker **pulses in time with its own drip**, so you can *see* the rhythm: a fast emitter flashes more often than a slow one. (Rates always lock to the BPM, or the host's tempo inside a DAW.)
- The emitter menu **stays open while you pick**, so you can audition rates against the pond and tap **Close** when it feels right.
- **New emitter rate** (drawer) sets the rate a *freshly placed* emitter starts at, from the same straight/dotted/triplet menu; existing ones keep whatever you set them to.
- **Each emitter can send on its own MIDI channel.** In the same hold menu, pick the emitter's **output channel** (1–16); the drawer sets the default a new emitter starts on (channel 1 to begin with). Point a few emitters at different channels and each drives a different synth, so several rates weave a multitimbral polyrhythm across your rig. This is separate from the Auto drip / MIDI trigger switch below: that chooses how emitters are *triggered*, this chooses where their notes are *sent* (an emitter played by MIDI still sends on its own channel). Wave interference always plays on channel 1.
- A freshly planted emitter **eases in**: its first drip waits about half a bar (at the current tempo) so it doesn't double up with the tap that placed it, then it settles into its rate.
- Up to **8 emitters**. The ring is its own control: **hold** it to open its menu (rate, channel, and **Remove**), and **drag** to move it around the pond (you can even

drag several at once, one per finger). A touch on the ring never drops a stray ripple.

- **Clear** removes them all at once. There's also a **quick clear-emitters button** (the slashed-circle, just under the paintbrush top-right) that appears whenever the pond has emitters, one tap clears them without opening the drawer.
- **Play the emitters over MIDI.** In the drawer's Emitters section, switch the trigger from **Auto drip** to **MIDI trigger** (the **Base note** control takes the place of *New emitter rate* while it's on). The emitters then stop auto-dripping and instead fire from incoming notes: emitter 1 plays on the **Base note** you set, emitter 2 a semitone up, and so on for all eight. Play them live from a controller in the app, or sequence them from your DAW when Pond is a plugin, so the pond keeps someone else's rhythm. Notes on any channel count. (If you route Pond's own output back into its input, MIDI mode can feed back, so keep that loop open, as with any MIDI instrument.)
- Fast rates with several emitters will use up all 32 simultaneous waves, extra taps wait for old waves to fade. That's the pond being honest about its physics.

Scale

- **Scale** (fifteen to choose from, the seven diatonic modes plus pentatonics, blues, whole tone, chromatic, harmonic and melodic minor, and the Japanese Hirajoshi) and **Key**, these feed **Generate**.
- **Transpose**, shift every note already in the pond up or down right now: ±1 semitone or ±12 (an octave). Unlike Key (which only colours the next Generate), this re-pitches what's there. Notes pushed past the top or bottom of the range clamp and may not come back on the way down, but **Undo** has you covered if a shift goes too far.
- **Octave range**, the Low/High steppers bound every note the pond *generates*: set C3–B5 for a comfortable middle register, or go wide for drama. Generate never leaves this range. Changing it doesn't touch notes already in the pond, they keep the pitch they were given; the range only shapes the *next* Generate.
- **Generate** deals a completely fresh pond: a random scatter of in-scale notes within the octave range, independent of what was there before. This is how you fill the pond. Tap it again for another roll.

- **Density** decides how sparse that scatter is. It's a smooth, **continuous** control reaching from a whisper-sparse **0.1%** all the way to a solid **100%**. The travel is heavily weighted to the sparse end: most of the slider explores **0.1–10%** (where the music is), and only the last stretch climbs on to a full grid, so fine control lands where you actually want it. It reads out to one decimal (e.g. *0.4%*, *3.7%*); a pond at 3–6% breathes, below 1% you get just a handful of notes, and near the top it fills right up. (Each Generate deals a fresh ~Density% of the grid, every time, it never compounds.)
- **Clear** silences every cell (waves still travel; they just find nothing to play).

Physics

How the waves behave and how each note sounds. The two toggles default off.

- **Wall reflections**, every wave bounces back off the pond's edges: drop near a wall and the echo sweeps back across, re-playing cells on the return trip, quieter with every extra cell of travel (the reflection's path length feeds the falloff). One tap becomes a conversation between a wave and its echoes. Costs wave slots: each drop uses up to five of the thirty-two, so go easy on emitters while it's on.
- **Wave interference**, turns ripples into *real* waves. Overlapping waves **add** where their crests meet and **cancel** where a crest meets a trough, just like rings from two stones in a real pond. A lone wave plays at a comfortable mid-level; reinforcing waves push a note **louder** (all the way to full velocity when several stack up), and cancelling waves push it **softer** (down to a whisper). It takes a lot of interference to reach either extreme. The grey ripples show the wave itself; on top, **blue** marks where waves reinforce and **yellow** where they cancel, so you can see the dynamics as well as hear them. Beautiful with two or three emitters, or a flurry of taps. *(Per-cell velocities you painted aren't used while this is on, interference is the dynamic.)*
    - **Velocity** (Base, Min, Max), shown while interference is on, is your velocity control for this mode. The interference decides where each note lands in the range: a lone ripple plays at **Base**, fully reinforcing waves rise toward **Max**, fully cancelling waves fall toward **Min**. Set them all equal for a flat, even pond; spread Min and Max wide for dramatic swells and near-silent troughs.

(This is the physics-based way Pond gives you velocity, rather than a fixed value per note.)

- **Falloff** (note shape), how quickly notes get quieter the farther they are from the splash point; far enough out, they fade to **silence**, so high falloff turns the pond into a spotlight around your touch. Gentle through most of its travel, tightening near the top.
- **Sustain** (note shape), note length, 1–1000 ms. Short = plucks and rain; long = washes and pads.

Sound

- **Built-in sound**, Pond's own voice, a soft 32-voice pluck, with its own volume. It's really a discovery aid so the app makes sound the moment you open it, no setup; turn it off when an external synth or DAW is making the sound.

About

- At the very bottom of the drawer: Pond's **version**, a **User Manual** button that opens this guide in the app, and **What's New**, the notes for each update (they also pop up once, the first time you open Pond after an update). In the app there's **Show intro again** too, which replays the first-launch tour and then puts your pond back exactly as it was.

---

5. Pond and MIDI

Pond speaks MIDI everywhere:

- **Standalone**, it publishes a CoreMIDI source named **"Pond"** that any app or hardware can subscribe to (it stays alive in the background). Notes go out with a computed velocity, and note-offs follow after the sustain time. Everything plays on **MIDI channel 1** until you give an emitter its own channel; then that emitter's notes go out on it, so one pond can drive several synths (see *Emitters*).

- **Send to devices**, open the drawer's MIDI section and use the **Send to** dropdown to pick any connected MIDI destinations (a USB interface, a class-compliant USB synth like a Digitone, a Eurorack bridge like the Shuttle Control, or another app's input). Pond sends its notes, and its clock, if "Send MIDI clock" is on, straight to the ones you tick. You can pick several at once. The list updates as you plug things in, and your choice is remembered. The **"Pond"** app port above is always published too, so app-to-app routing keeps working alongside any devices you select.
- **MIDI clock** flows in or out, see the MIDI section.

Pond as an AUv3 plugin

Inside AUM, Logic, Cubasis and friends, Pond loads as an **Audio Unit MIDI processor** ("naturarum: Pond"), the full pond UI in a plugin window. The plugin window is **freely resizable**: the whole pond scales to fit whatever size and shape you give it (square cells, centred), so it never crops or spills, resize it as small or large as you like.

Pond **follows the host transport**: when you **stop** the host, the emitters stop dripping (and pick back up on **play**), so the pond sits in time with the rest of your session. Notes you tap by hand still play while stopped, and the tempo always comes from the host.

In **AUM**: add a **MIDI channel** → load Pond in its slot (under *Audio Unit MIDI Processor*) → on your synth's node, open its MIDI input panel and enable **the MIDI channel containing Pond** (it's listed by the channel's name, not "Pond"). Done, Pond plays your synth.

In a host, Pond **follows the host tempo** automatically (the BPM readout shows "· HOST") and quantize boundaries land on the host's actual beats. Your session saves Pond's entire state, and the host's own **user-preset menu** works too. Pond's own **patterns are all there as well**, the same library as the app (see Presets above), with the bottom-left switcher for hot-swapping mid-session.

---

6. Recipes

- **Generative chill**: Pentatonic Major, Generate at ~4% density, one emitter at 1 bar, quantize ⅛, sustain ~400 ms. Walk away. It plays better than it has any right to.
- **Polyrhythmic web**: two emitters far apart, drip rate ½, quantize 1/16, high falloff. The interference pattern is the rhythm.
- **Performance surface**: no emitters, quantize off, short sustain, a denser Generate (~15%). Pure hand-drumming on water.
- **Patch DJ**: fill a page of pattern slots with contrasting ponds, drag them into order, and step through them with the **‹ name ›** switcher on the pond, waves in flight survive the swap and rake across the incoming notes.
- **One pond, three synths**: three emitters at 1 bar, ½ and ⅛T, each on its own MIDI channel, each channel going to a different synth. A polyrhythm spread across your whole rig.
- **Echo chamber**: reflections on, quantize ⅛, ONE tap near a corner, high falloff. The wave and its four echoes answer each other in decaying phrases, Pond's version of a delay pedal.

---

7. If something's quiet

- **Fresh install?** The pond starts empty on purpose, Generate or paint some notes first.
- **Nothing moving?** Check the **pause** button (top-right) isn't showing a ▸ play glyph, tap it to resume.
- **No sound, standalone?** Check the Built-in sound toggle (the Sound section near the bottom of the drawer), then the iPad mute switch / volume.
- **No sound from the AU?** It makes no sound by itself, it sends MIDI. Check the routing recipe in section 5.
- **Bluetooth MIDI dropping notes on a busy pond?** A Bluetooth MIDI link carries far less than a cable. If a very dense pond sends more than it can keep

up with, Pond eases off on new notes rather than letting the link fall behind (note-offs always get through, so nothing hangs on your gear, and the built-in sound still plays everything). A lighter Density, fewer emitters, or a cable gets every note through.

- **USB / hardware synth silent?** Pond only sends to devices you pick: open the drawer's **Send to** dropdown (MIDI section) and tick your device. If it isn't listed, reseat the cable, the list refreshes on its own when the connection changes.
- **Pond missing from the host's plugin list?** Force-quit the host and reopen; after an update, reboot the iPad if needed (iOS can be slow to register fresh plugins).
- **Clock sync says "Waiting..."?** Pond ignores its own clock output by design, make sure another device or app is actually sending clock to the iPad.
- **Taps not making waves?** If several emitters are dripping fast, all 32 wave slots may be busy, clear an emitter or slow the drip.